Floor, Ceiling & Discrete Functions - Assimilation Book

For a Grade-9 learner who may have seen the symbols but does not yet reliably control intervals, endpoints or negative inputs.

1. RECONNECT - what the symbol is trying to say

You may remember that floor(3.7)=3. That is not a rule about deleting decimals. The real definition is:

$\text{floor}(x)=n \iff n \le x < n+1$, for an integer n.

Likewise,

$\text{ceil}(x)=n \iff n-1 < x \le n$.

The two symbols encode half-open intervals.

Try before reading further:

- What is floor(-2.3)?
- What is ceil(-2.3)?
- Solve floor(x)=4 without writing x=4.

2. DISCOVER - the staircase interval

If floor(x)=4, then every real number from 4 up to, but not including, 5 works:

x in [4,5).

If ceil(x)=4, then

x in (3,4].

The endpoint direction is the first major decision boundary:

- floor: include the lower integer, exclude the next;
- ceiling: exclude the previous integer, include the upper integer.

3. MAKE SENSE - negative inputs expose truncation errors

$-3 < -2.3 < -2$.

The greatest integer not exceeding -2.3 is -3, so

floor(-2.3)=-3.

Truncating toward zero would give -2, which is wrong for floor.

For ceiling, the least integer at least -2.3 is -2:

ceil(-2.3)=-2.

So the safest habit is not "drop decimals". It is "locate the number between consecutive integers".

4. Integer shifts and reflection

For every integer k,

floor(x+k)=floor(x)+k,

ceil(x+k)=ceil(x)+k.

Also,

ceil(x)=-floor(-x).

These are representation shortcuts, not new casework.

5. Fractional part

Define

$\{x\}=x-\mathrm{floor}(x)$.

Then always

$0<=\{x\}<1$.

For positive decimals this resembles the digits after the point. For negative numbers it does not:

$\{-2.3\}=-2.3-(-3)=0.7$.

That is why "fractional part" is structural, not typographical.

6. Floor equations are interval equations

Example:

$\mathrm{floor}((2x+1)/3)=4$.

Do not write $(2x+1)/3=4$.

Translate first:

$4 <= (2x+1)/3 < 5$.

Then solve:

$11/2 <= x < 7$.

One discrete value has encoded a whole real interval.

For ceiling,

$\mathrm{ceil}((3x-1)/2)=5$

means

$4 < (3x-1)/2 <= 5$,

so

$3 < x <= 11/3$.

7. Real interval first, integer filter second

Suppose n is required to be an integer and

$\mathrm{floor}((n+1)/3)=2$.

First solve over the reals:

$2 <= (n+1)/3 < 3$,

hence

$5 <= n < 8$.

Now use the integer filter:

n in $\{5,6,7\}$.

Do not mix the two stages mentally. First decode the real interval; then intersect it with the required discrete domain.

8. Counting integers in a real interval

If you need integers m with

a <= m < b,

the first candidate is ceil(a) and the last is ceil(b)-1. The count is

ceil(b)-ceil(a)

when the interval is nonempty.

For a closed interval [a,b], the count is

floor(b)-ceil(a)+1

when nonempty.

9. TRY - attempt before help

Track A - floor equation

Solve floor((x-1)/2)=3.

Attempt first.

- Full support: write 3 <= (x-1)/2 < 4, then solve both sides.
- Structural prompt: replace the floor equation by its half-open interval.
- Recognition prompt: which endpoint is strict for floor?
- Independent target: solve floor((3x+2)/5)=-1 independently.

Track B - ceiling equation

Solve ceil((x+2)/3)=2.

Attempt first.

- Full support: write 1 < (x+2)/3 <= 2.
- Structural prompt: ceiling uses a left-open, right-closed interval.
- Recognition prompt: decode before doing algebra.
- Independent target: solve ceil((2x-5)/4)=-2 independently.

Track C - integer filtering

Find all integers n with floor((n-2)/4)=3.

Attempt first.

- Full support: 3 <= (n-2)/4 < 4, then keep only integers.
- Structural prompt: solve a real interval, then intersect with Z.
- Recognition prompt: do not count before the interval is correct.
- Independent target: find the number of integers n satisfying ceil(n/5)=3.

10. DIAGNOSE - common wrong starts

Wrong start 1: floor means truncation

Works accidentally for some positive numbers; fails immediately for negative inputs.

Wrong start 2: floor(f(x))=k means f(x)=k

False. The correct statement is k<=f(x)<k+1.

Wrong start 3: forgetting the strict endpoint

Changing [k,k+1) into [k,k+1] can create a false extra solution.

Wrong start 4: filtering integers too early

First solve the real interval. Then apply the integer condition.

Wrong start 5: importing general inequality machinery

Most floor problems are not optimization problems. The first move is usually interval translation, not an inequality theorem.

11. ADOPT - seven mental rules

1. floor(x)=n <=> n<=x<n+1.
2. ceil(x)=n <=> n-1<x<=n.
3. Negative input: never trust truncation intuition.
4. Integer shifts pass through floor/ceiling.
5. ceil(x)=-floor(-x).
6. {x}=x-floor(x) lies in [0,1).
7. Decode interval first; integer-filter second.

12. Validated IOQM anchors

- IOQM-2024-Q21: two floor intervals intersect digit structure; answer 91.
- IOQM-2024-Q26: set n=floor(x), use x in [n,n+1), and test feasibility; floor values 16 and 17 sum to 33.

Use the validated paper for exact historical wording.

13. TRANSFER

The same mechanism appears when:

- a counting problem asks how many integer labels lie inside a real interval;
- a number-theory condition narrows an integer into a one-unit floor interval;
- a combinatorics state is selected by floor(t/T);
- a negative parameter makes truncation and floor diverge.

The surface changes. The first question stays:

> "What half-open interval does this discrete value encode?"

ALG-07 - First-Step Reference

Use after the Assimilation Book.

Recognition atlas

Visible clue	First thought
floor(f(x))=n	n<=f(x)<n+1
ceil(f(x))=n	n-1<f(x)<=n
negative decimal inside floor	locate between consecutive integers; do not truncate
integer shift x+k	pull integer k outside floor/ceiling
fractional part	x-floor(x)
real interval + integer target	solve real interval, then intersect with Z
count integers in [a,b)	first ceil(a), last ceil(b)-1
floor(x)=ceil(x)	test whether x must be an integer

Quick router

text
FLOOR or CEILING VALUE GIVEN?
 -> write the correct half-open interval

NEGATIVE INPUT?

-> use order, not truncation

INTEGER SHIFT?
-> use translation identity

REAL INTERVAL FOUND?
-> if target is integer, filter only now

COUNTING INTEGERS?
-> identify first/last admissible integer and check endpoints

First-step cards

Floor equation

floor(A)=n -> n<=A<n+1.

Ceiling equation

ceil(A)=n -> n-1<A<=n.

Negative input

Find the consecutive integers surrounding the input.

Fractional part

Write {x}=x-floor(x) before looking at decimal digits.

Integer filter

Solve the real interval, then intersect with Z.

Contrast strip

- floor vs truncation: negative inputs expose the difference;
- floor vs ceiling: opposite half-open endpoint convention;
- floor equation vs ordinary equation: interval vs single equality;
- real interval vs integer solutions: continuum vs discrete filter;
- included vs excluded endpoint: brackets are mathematical data, not typography.

30-second self-check

Before calculating, ask:

1. What integer is the floor/ceiling claiming?
2. Which half-open interval corresponds to it?
3. Which endpoint is strict?
4. Is the final variable real or integer?
5. Can a shift/reflection identity shorten the work?

ALG-07 - Recognition and First-Line Lab

Write only the first useful mathematical line unless the item explicitly asks for a value.

1. Evaluate floor(3.8) by locating 3.8 between consecutive integers.
2. Evaluate floor(-3.8) without truncation.
3. Evaluate ceil(-3.8).
4. floor(x)=5. Write the encoded interval.
5. ceil(x)=-2. Write the encoded interval.
6. floor((2x+1)/3)=4. Write the first inequality chain.
7. floor(x)<3. Write an equivalent ordinary inequality.
8. When can floor(x)=x?
9. Write the defining first line for the fractional part of -2.3.
10. Count integers in [-2.4,3.1) by writing the first and last candidate.
11. Simplify floor(x+7) without casework.
12. Express ceil(x) using floor and reflection.

13. floor(x)=2 and x is an integer. Write the interval before filtering.
14. floor(x^2)=3. Write the first double inequality.
15. floor(x)=ceil(x). State the structural condition you should test.
16. For floor(x)=4, identify the included endpoint and excluded endpoint.

No method-control labels should appear in later independent mastery items.

ALG-07 - Practice and Transfer Bank

All items are author-created.

F0 - Foundation

1. Evaluate floor(4.6), floor(-4.6), ceil(4.6), ceil(-4.6).
2. Translate floor(x)=n and ceil(x)=n into inequalities.
3. Find {3.25} and {-2.3}.
4. Verify ceil(x)=-floor(-x) for x=2.4 and x=-2.4.

F1 - Direct

5. Solve floor(x)=4 over the reals.
6. Solve ceil(x)=4 over the reals.
7. Solve floor((x+2)/3)=2.
8. Solve floor(-x)=3.
9. If 3<x<4 and {x}=0.25, find x.

F2 - Standard

10. Solve floor(2x+1)=5.
11. Solve ceil(3x-2)=4.
12. Solve floor(x)=ceil(x).
13. Solve floor(x)+ceil(x)=7.
14. Classify the value of floor(x)+floor(-x) for integer and non-integer x.
15. Solve floor(x^2)=4.

F3 - Disguised

16. How many integers lie in [-3.7,5.2)?
17. How many integers k satisfy floor((k+1)/3)=2?
18. Solve simultaneously floor(x)=2 and floor(2x)=5.
19. Explain why ceil(x)=-floor(-x) follows from the two interval definitions.
20. Characterize all real x such that floor(x+1/2)=floor(x).

F4 - Preliminary-style transfer

21. Can floor((x-1)/2)=floor((x+1)/2) ever hold? Prove your answer without case enumeration.
22. Solve floor(x)+floor(x+1/2)=5.
23. Find x if floor(x)=3 and {x}=0.7.
24. If floor(x)=-4, state the full possible interval for x and explain why truncation can mislead.
25. Find all integers n satisfying floor(n/4)=3.
26. How many integers n satisfy ceil(n/5)=3?
27. For x>0, solve floor(1/x)=2.
28. Prove floor(x+k)=floor(x)+k for integer k from the interval definition.
29. WHY-NOT: why does floor(x)=2 not allow the replacement x=2?
30. A gate is active for real times t in [2.3,8.0). How many integer timestamps occur while it is active?

ALG-07 - Independent Mastery Check

No default hints. No method labels.

Part A - Notice and first useful line

1. floor((3x-2)/4)=5. Write only the first inequality chain.
2. ceil((x+3)/2)=-1. Write only the first inequality chain.
3. floor(x^2)=7. Write only the first double inequality.

4. How many integers n satisfy -4.2<=n<3.1? Write only the first and last possible integer.

Part B - Full mixed solve

5. Solve floor((2x+1)/3)=4.
6. Solve ceil((3x-1)/2)=5.
7. Solve floor(x)+ceil(x)=-5.
8. Solve floor(x^2)=9.
9. Find all integers k satisfying floor((k-2)/4)=3.
10. Find the fractional part of -2.75.
11. Solve floor(x)=ceil(x).
12. Characterize all real x satisfying floor(x+0.4)=floor(x).

Part C - Same surface, different decision

13. Solve both floor(x)=4 and ceil(x)=4, then explain why the endpoints differ.
14. Compare floor(-2.3) with truncation of -2.3 toward zero. Explain which definition controls the answer.

Part D - Transfer and WHY-NOT

15. Seat labels are integers. How many labels lie in the real interval [12.4,20.0]?
16. A learner solves floor(2x)=3 by writing 2x=3. Explain the defect and give the complete solution set.

Mastery reflection

For any three solved items, add:
- What interval did the discrete value encode?
- Which endpoint was strict?
- Did the final domain require an integer filter?
- What nearby wrong move was tempting?
- What final endpoint/domain check did you perform?